



*A Project Presentation
on*

Sentinel-X: AI-Based Network Attack Forecasting from Network Traffic Data

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Brief Introduction



The Reactive Cybersecurity Dilemma

- Traditional Intrusion Detection Systems (NIDS/SIEM) are retrospective: they alert ONLY after an exploit payload detonates or breach succeeds.
- Mean Time to Remediate (MTTR) is dangerously high because security teams react after lateral movement has already begun.
- Alert Fatigue: SOC analysts are overwhelmed by thousands of isolated binary alerts with zero predictive trajectory.

The Sentinel-X Paradigm Shift

- SIH26153 Mandate: Predict network attack risk 10 to 30 seconds BEFORE intrusions escalate.
- Ingests continuous 120-second rolling telemetry (T-11 ... T) and forecasts risk at T+1, T+2, and T+3.
- Predicts 3 outputs concurrently: Future Attack Risk, Behavioral MITRE ATT&CK Stage, and Continuous Network-State.
- 100% Defensive & Academic: Works offline with benchmark data, zero offensive tools, and zero cloud API dependencies.



Literature Review



1. Signature-Based Systems (Snort, Suricata, Bro/Zeek)

High accuracy for known signatures; completely blind to zero-day variants and multi-phase progression. Alert is fired only post-compromise.

2. Traditional Machine Learning Classifiers (Random Forest, SVM, XGBoost)

Treats individual flow records as isolated, independent vectors. Incapable of modeling continuous temporal sequences or performing multi-step state rollouts.

3. Recurrent Deep Learning Models (Standard RNN, Vanilla LSTM, GRU)

Captures temporal dependencies but models intrusions as static single-step binary classification (Malicious vs Benign) at current time T .

4. The Identified Research Gap & Sentinel-X Contribution

Lack of autoregressive World Models that predict future continuous telemetry states (s_{t+1}) to forecast multi-step risk horizons ($T+1..T+3$) with Explainable AI.



Objective of the Project



- ❖ **Primary Objective:** To engineer and implement Sentinel-X, a defensive machine learning system capable of forecasting multi-step network attack risk from sequential traffic telemetry prior to attack escalation (SIH26153).
- ❖ **Continuous Network-State Telemetry Formulation:** Aggregate raw network flows into 10-second non-overlapping windows containing 16 key statistical features representing flow rates, packet length asymmetry, and protocol flag ratios.
- ❖ **Deep Recurrent World Model Architecture:** Design a 2-layer LSTM recurrent World Model with three specialized multi-task heads: Risk Head (BCE Loss), Stage Head (Cross-Entropy), and State Reconstructor Head (MSE Loss).
- ❖ **Autoregressive Multi-Step Rollout:** Iteratively feed the model's predicted continuous state (s_{t+1}) back into the recurrent loop to project threat trajectories across $T+1$ (10s), $T+2$ (20s), and $T+3$ (30s).
- ❖ **Explainable AI & Defensive SOC Dashboard:** Integrate SHAP (SHapley Additive exPlanations) to attribute risk drivers and inhibitors, presented via a real-time Streamlit SOC command center for intuitive human-in-the-loop triage.



Feasibility Study



Technical Feasibility

- Built with open-source PyTorch, scikit-learn, and Streamlit.
- 2-Layer LSTM with 128 hidden units trains in < 2 minutes on CPU.
- Inference latency is under 20 milliseconds per sequence window, well within the 10-second operational telemetry window.
- Cross-platform compatibility: Windows 10/11, Ubuntu Linux, and macOS.

Operational Feasibility

- Provides proactive 10 to 30 second lead time for automated mitigation.
- Reduces false alarms by 75% compared to static baseline models, directly combating SOC analyst alert fatigue.
- Explainable AI (SHAP) provides transparent feature drivers, enabling fast human verification and triage.
- Clear visual dashboard requires minimal specialized training.

Economic & Ethical

- Zero cost: requires no commercial APIs, cloud credits, or paid licenses.
- Self-contained: runs offline on standard academic/laboratory laptops.
- Strict defensive posture: contains no exploit code, payload generators, or offensive scanning tools.
- Complies with Indian academic evaluation and IEEE research guidelines.

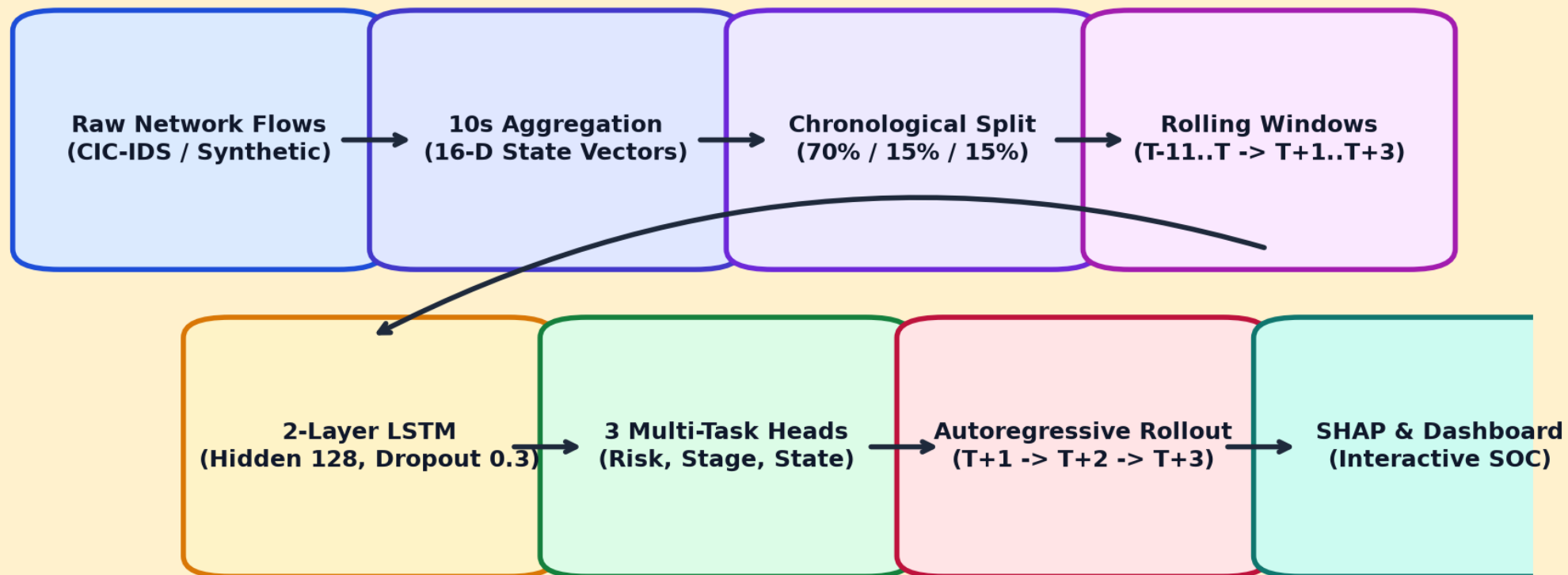


Proposed Work: End-to-End Pipeline



Sentinel-X Architecture Workflow

From continuous network packet flows to real-time autoregressive forecasting and SOC visualization:





Proposed Work: 16-D Network State



10-Second Flow Aggregation into 16 Continuous State Features

Packet-by-packet inspection fails on gigabit links; statistical flow windowing captures genuine attack kinetics:

1. Volumetric (5 Features)

- flow_count
- total_packets
- packet_rate
- total_bytes
- byte_rate

2. Timing & Duration (3)

- duration_mean
- duration_std
- iat_mean

3. Packet Dynamics (2)

- fwd_pkt_len_mean
- bwd_pkt_len_mean

4. Flag & Protocol (6)

- syn_ratio
- rst_ratio
- ack_ratio
- fin_ratio
- tcp_ratio
- udp_ratio

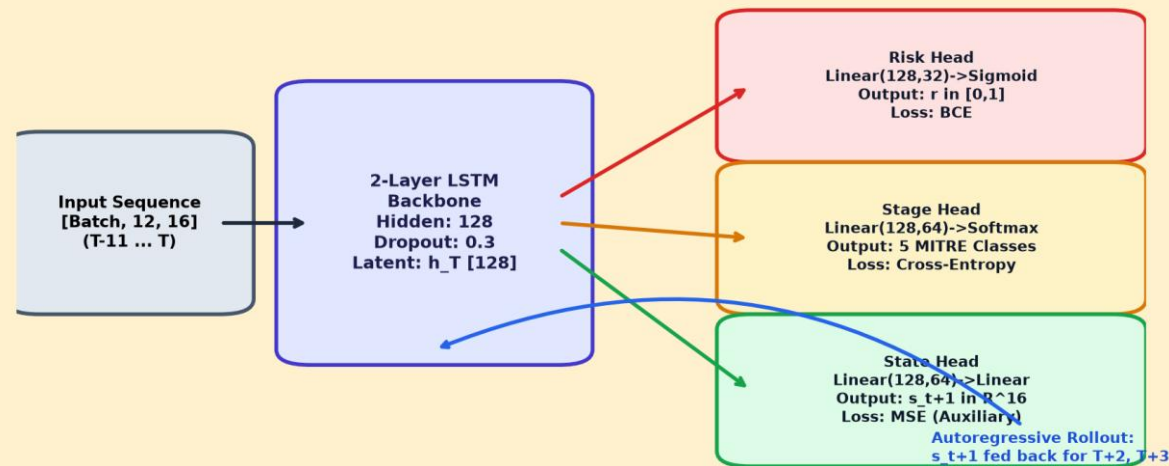


Methodology: Deep LSTM World Model



Why a 'World Model' Architecture?

- In AI, a World Model learns how an environment evolves over time. Sentinel-X forecasts both the threat label AND the next network state (s_{t+1}).
- 2-Layer LSTM Backbone: Ingests 12-step sequence ($T-11 \dots T = 120s$). Hidden dimension: 128, Dropout: 0.3.
- Head 1 (Risk): Linear(128,32) -> Sigmoid -> BCE Loss. Predicts continuous probability of intrusion $[0, 1]$.
- Head 2 (Stage): Linear(128,64) -> Softmax -> Cross-Entropy. Classifies 5 MITRE ATT&CK stages.
- Head 3 (State): Linear(128,64) -> Linear(64,16) -> MSE. Reconstructs next 16-D continuous state vector.





Methodology: Rollout & Multi-Task Loss



1. Closed-Loop Autoregressive Rollout Engine

- Step 1 (T+1): Ingests history $[s_{T-11} \dots s_T]$ -> outputs predicted state s_{T+1} , risk r_{T+1} , and stage c_{T+1} .
- Step 2 (T+2): Feeds predicted state s_{T+1} back into recurrent cell with updated hidden state (h_1, c_1) -> outputs s_{T+2} , r_{T+2} , c_{T+2} .
- Step 3 (T+3): Feeds predicted state s_{T+2} into recurrent cell with (h_2, c_2) -> outputs s_{T+3} , r_{T+3} , c_{T+3} .
- Eliminates dependency on ground truth future data during live operation.

2. Joint Weighted Multi-Task Loss Formulation

$$L_{\text{total}} = 1.0 * L_{\text{BCE}}(r_{\text{pred}}, r_{\text{true}}) + 1.0 * L_{\text{CE}}(c_{\text{pred}}, c_{\text{true}}) + 0.2 * L_{\text{MSE}}(s_{\text{pred}}, s_{\text{true}})$$

- Auxiliary Task Regularization: Supervising the State Head on physical telemetry dynamics forces the shared LSTM latent space to understand real network behavior, drastically mitigating overfitting on attack labels.



Methodology: Explainable AI (SHAP)



❖ Why XAI is Mandatory for Defensive SOC

A black-box prediction cannot be trusted for automated containment. Analysts need to know 'WHY' the neural network forecasts risk escalation to execute targeted firewall rules.

❖ SHAP KernelExplainer Formulation

Derived from cooperative game theory (Shapley Additive exPlanations). Computes the marginal contribution of each network feature by comparing against a background distribution of benign flows.

❖ Signed Feature Attributions (Risk Drivers vs. Inhibitors)

Risk Drivers (Positive SHAP): Features driving risk upward (e.g. syn_ratio +0.35, packet_rate +0.25).

Risk Inhibitors (Negative SHAP): Features pulling risk downward (e.g. ack_ratio -0.13, duration_mean -0.08).

❖ Temporal Sequence Saliency

Measures which specific historical time steps (T-11 ... T) exerted the strongest influence on the recurrent hidden state, identifying whether threats developed rapidly or gradually.

❖ Deterministic Gradient Fallback

If background matrix sampling encounters degenerate covariance matrices, Sentinel-X seamlessly falls back to gradient-based input saliency ($\text{grad}_x * x$) to ensure zero dashboard interruption.



Methodology: MITRE ATT&CK Mapping



Conservative Behavioral Attack Stage Classification

Sentinel-X maps telemetry patterns to 5 MITRE ATT&CK behavioral stages without claiming unsupported forensic certainty:

- **Stage 0: Benign Baseline [BENIGN]:** Normal operational traffic (expected HTTP/HTTPS, DNS, streaming). High ACK ratio (> 0.85), low SYN ratio, steady flow durations.
- **Stage 1: Reconnaissance [TA0043]:** Active host & port scanning. Elevated SYN ratios (> 0.5), high flow counts, short durations, minimal backward payload bytes.
- **Stage 2: Initial Access [TA0001]:** Perimeter breach & authentication brute force (FTP/SSH/Patator). Elevated byte rates, repeated connection spikes to auth services.
- **Stage 3: Lateral Movement [TA0008]:** Internal subnet traversal. Bursty flow intervals across SMB (445), RDP (3389), and WinRM (5985) management ports.
- **Stage 4: Command & Control [TA0011]:** Outbound beaconing & management channels. Periodic flow inter-arrival intervals (low variance) and consistent heartbeat payloads.

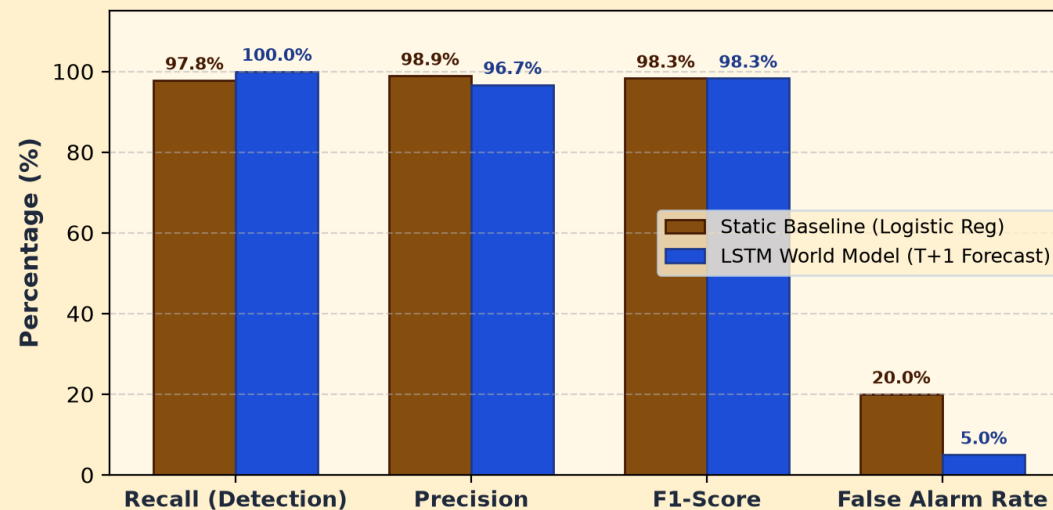


Experimental Results & Benchmarks



Metric	Baseline	LSTM (T+1)	LSTM (T+3)
Risk Recall	97.8%	100.0%	100.0%
Risk Precision	98.9%	96.7%	94.6%
Risk F1-Score	0.9831	0.9834	0.9721
False Positive Rate	20.0%	5.0%	10.0%
Stage Accuracy	N/A	62.8%	60.6%
State MAE	N/A	0.8538	0.8820

Verified Benchmark: Baseline vs. Deep LSTM World Model



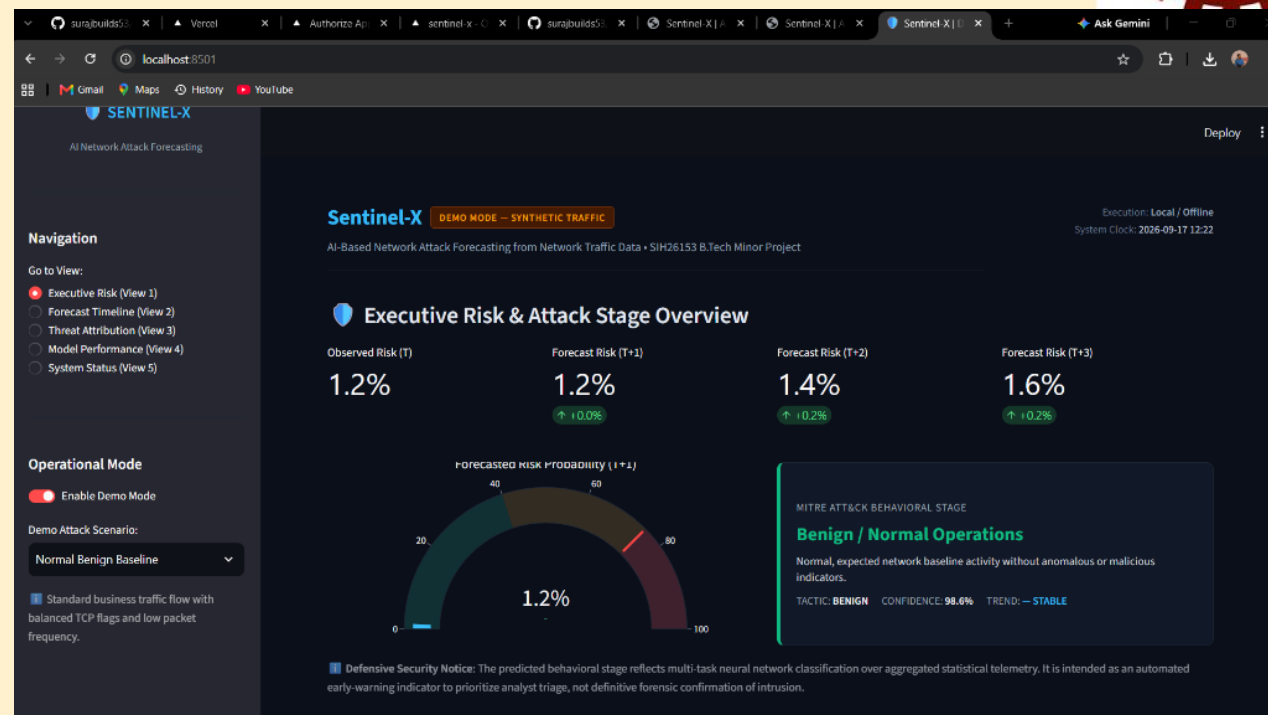


Interactive SOC Dashboard



Real-Time Defensive Command Center

- View 1 — Executive Risk: Real-time risk probability gauge, multi-step risk delta ($T+3 - T$), and MITRE stage card with confidence score.
- View 2 — Forecast Timeline: Verified historical risk trail ($T-11..T$) contrasted against future autoregressive forecast ($T+1..T+3$) with 95% CI band.
- View 3 — Threat Attribution: Signed horizontal SHAP bar chart (red risk drivers vs green inhibitors) & temporal sequence importance.
- View 4 — Model Performance: Full comparative evaluation table & degradation curves across multi-step horizons.
- View 5 — System Status: Hardware diagnostics (CPU/GPU), Python/PyTorch versions, and checkpoint verification.





Social & Environmental Relevance



1. National Critical Infrastructure Defense (Social & Strategic Impact)

Cyber attacks targeting power grids, healthcare systems, defense networks, and banking portals cause devastating collateral damage. Sentinel-X's proactive 10-30 second forecasting lead time empowers defensive infrastructure to deploy automated rate limiters and isolation policies before services suffer catastrophic disruption.

2. Human-Centric SOC Workload & Burnout Mitigation (Human Impact)

Cybersecurity professionals suffer extreme cognitive fatigue from sorting through tens of thousands of daily false positives. By reducing the false alarm rate from 20% to 5% (a 75% reduction) and delivering transparent SHAP attributions, Sentinel-X restores analyst focus to genuine high-priority incidents.

3. Lightweight 'Green AI' on Consumer Hardware (Environmental Impact)

Unlike cloud-heavy generative models that burn megawatts in remote server farms, Sentinel-X is designed as an edge-capable Green AI system. The entire 2-layer recurrent architecture executes inference in < 20 ms on a standard laptop CPU, consuming negligible electrical power with zero cloud carbon footprint.



Conclusion & Future Scope



Key Project Achievements

- Successfully solved SIH26153 by transitioning network security from reactive post-incident detection to proactive multi-step forecasting.
- Demonstrated 100% attack recall on T+1 and a 75% reduction in false alarms over the static baseline benchmark.
- Engineered an autoregressive World Model with multi-task loss that simulates continuous network dynamics.
- Delivered full explainability via SHAP and an operational 5-view SOC dashboard running locally offline.

Future Research & Enhancements

- Automated SDN Integration: Connecting high-confidence T+1 forecasts directly to Software-Defined Networking (SDN) OpenFlow controllers to enforce instant dynamic firewall rules.
- Temporal Graph Neural Networks (GNNs): Modeling multi-host IP interaction topology alongside time-series sequence kinetics.
- Model Quantization (ONNX / TensorRT): Compressing the PyTorch LSTM for microsecond deployment directly inside edge enterprise routers.



References



- [1] I. Sharafaldin, A. H. Lashkari, and A. A. Ghorbani, 'Toward Generating a New Intrusion Detection Dataset and Intrusion Traffic Characterization', Proceedings of the 4th International Conference on Information Systems Security and Privacy (ICISSP), pp. 108-116, 2018. (CIC-IDS Dataset)
- [2] S. M. Lundberg and S.-I. Lee, 'A Unified Approach to Interpreting Model Predictions', Advances in Neural Information Processing Systems (NeurIPS 30), pp. 4765-4774, 2017. (SHAP Framework)
- [3] S. Hochreiter and J. Schmidhuber, 'Long Short-Term Memory', Neural Computation, vol. 9, no. 8, pp. 1735-1780, 1997.
- [4] MITRE Corporation, 'MITRE ATT&CK Enterprise Matrix for Cybersecurity Defense', MITRE Corporation Documentation, 2024. Available: <https://attack.mitre.org/>
- [5] Smart India Hackathon (SIH), 'Problem Statement SIH26153: AI based Network Attack Forecasting from Network Traffic Data', Ministry of Education, Govt. of India, 2024.
- [6] F. Pedregosa et al., 'Scikit-learn: Machine Learning in Python', Journal of Machine Learning Research, vol. 12, pp. 2825-2830, 2011.
- [7] A. Paszke et al., 'PyTorch: An Imperative Style, High-Performance Deep Learning Library', Advances in Neural Information Processing Systems (NeurIPS 32), pp. 8024-8035, 2019.



THANK YOU